



Fig. 3: CarIQ in use

Connectivity in real-time

The platform is assisted by over-the-air network communication protocols such as SSL/HTTPS. On top of this, a VPN network and telecommunications provider has also been partnered with, for securing the communication channel. The device comes with GPS for location awareness and GSM for securely transmitting data to the Cloud.

transmitting information back to the hub. It goes one step further from on-board diagnostics on CarIQ platform.

Further, this system provides a different set of in-depth details about the car such as the capability to monitor the health summary, which the team claims could help save a lot of money and get the car geared up for maintenance. FleetIQ also makes sure that drivers are sent tips to improve their driving techniques, thereby assuring that the car is in good hands.

Smartcars are en route

Coming to interoperability in transformational integration, CarIQ could be termed as a completely autonomous platform. With its user-friendly trending application available on online application stores, you can experience its assistance on-the-go, wherever you are.

Creation of a tandem between the user and the car using the most intelligent components of such similar platforms is where the IoT emerges victorious.

"Since we have indigenously built our hardware, there is scope for expansion. Add-ons for hardware can be integrated with CarIQ devices to provide better experience to the user. These add-ons can communicate with CarIQ device using custom protocols. These can also make use of CarIQ's robust platform. Anyone can get in touch with us if they think they have a hardware product that is relevant," says Thomas.

With such path-breaking innovations around, let us begin by saying that the world is ready for intelligent cars of the future! **EFY**

At the heart of the platform

- Automotive Interpreter decodes vehicle language to data—the platform humans understand
- Rules Engine decides what to do with a data packet in real time
- Alerting Engine sends out real-time alerts based on Rules Engine
- Big Data Store is where all data goes
- Big Data Analytics Engine is where all data from Big Data Store is periodically run through the analytics engine to come out with more inferences and patterns

thousands of cars are involved.

"The hardware went through multiple iterations to get to the state it is in today. Developing the hardware was the biggest challenge. Prototyping labs take time, cost and patience. Turnaround time for getting the prototypes ready was another hurdle. Hardware development took time to reach a state that we were satisfied with," says Thomas.

The next challenge in integration would be developing the soul on which CarIQ runs. Software and its updates are the differentiators in any smartproduct we see these days. The features and additions are made friendly and accessible to users with simple smartphone application updates, without any hassles.

Arriving at accurate estimations

Leveraging the platform's highly-reliable components, accurate estimations of mileage, gear ratios, engine health, car health, driver rating and

drive scores are acquired. The team has developed a unique way in which data is collected and sent, thereby minimising bandwidth usage.

Also included is an accelerometer, onboard storage, Bluetooth and processors to control all this. The accelerometer aids in identifying rash-driving incidents and accidents while alerting engine monitors about such events before kicking in. Onboard storage is used when GSM network faces connectivity issues. This way drive data is never lost.

Big Data team takes care of architecture and advanced analytics on the data, while Microsoft Azure forms the very cradle on which this robust architecture rests. It performs the function of analysing cars of similar makes and models, and benchmarks these for analytics and scoring.

FleetIQ is one such extension to CarIQ. It features the fleets' requirements. FleetIQ can monitor and manage all cars through a single portal, leveraging minimal bandwidth for